Masked Autoencoders with Neural Cellular Automata for Spatio-spectral Reconstruction of PHANGS–ALMA Cubes

Mamadou Barry

October 2026

Abstract

This project investigates whether recurrent local refinement using a Neural Cellular Automaton (NCA) improves the reconstruction of masked PHANGS–ALMA CO(2–1) spectral cubes compared with a Conv3D refiner. The dataset comprised 68 galaxy cubes: 44 for training, 19 for validation, and five for independent testing. Within a common masked-autoencoder framework, the refiners were trained using a matched 40-epoch protocol and evaluated under random, frequency-varying, and structured masking. Evaluation included voxel-level metrics, such as PSNR, SSIM, signal-region MSE, and spectral-correlation MAE, alongside scientific diagnostics including moment-0 correlation and integrated-emission recovery. Across the five test galaxies, NCA configurations achieved lower signal-region MSE under random and frequency-varying masking. Under structured masking, the best Conv3D configuration slightly outperformed the best NCA configuration, although neither reconstructed the missing emission accurately; the highest structured normalized moment-0 recovery was only 41.90%. Improved voxel-level reconstruction did not consistently imply better astronomical-emission recovery, with Conv3D achieving the closest-to-complete frequency-varying normalized moment-0 recovery of 99.67%. With five independent galaxies, the smallest attainable exact two-sided Wilcoxon p-value was 0.0625. These findings should therefore be interpreted as directional and require confirmation using a larger independent test sample.

1 Introduction

Cold molecular gas provides the material from which stars form. Its distribution and density influence where stars emerge and how efficiently a galaxy converts gas into stars. Measuring molecular gas across different regions of a galaxy helps us understand star formation and galactic evolution. Whilst hydrogen is the dominant gas in molecular clouds, it is difficult to observe directly under cold conditions. Carbon monoxide (CO) emission is more readily observable and is therefore widely used as an indirect tracer of molecular gas. Converting CO intensity into an H_2 mass requires a CO-to-H_2 conversion factor and assumptions about excitation and environment [1].

The PHANGS–ALMA survey used CO(2–1) to map 90 nearby star-forming galaxies at approximately $1''$, corresponding to a typical physical resolution of roughly 100 pc [2]. This scale is comparable to the dimensions of giant molecular clouds and therefore connects galactic structure to cloud-scale gas properties [3]. CO(2–1) does not provide molecular mass without assumptions. Its intensity depends on factors including excitation, optical depth and the CO(2–1)/CO(1–0) line ratio. It is therefore best described as an observational tracer of molecular gas rather than a direct measurement of hydrogen [1]. We use masked cube reconstruction as a controlled setting for comparing NCA and Conv3D refinement, testing robustness to missing-data geometry, and examining whether reconstruction metrics track scientific observables such as integrated flux. Machine-learning models can learn recurring spatial and spectral relationships from a collection of cubes and use neighbouring spatial positions and spectral channels simultaneously when reconstructing masked regions. In this benchmark, a masked autoencoder can reconstruct cubes with 50% of patches hidden, and the NCA refiner improves reconstruction metrics relative to the tested Conv3D refiner under random and frequency-varying masking. However, the best voxel-level reconstruction metrics do not always imply integrated-flux recovery closest to 100%.

2 Related Work

2.1 Deep Learning in Radio Astronomy

Radio interferometers do not observe images directly; instead, they sample the Fourier plane incompletely [4]. Image reconstruction is therefore an inverse problem [4, 5]. Classical reconstruction methods such as CLEAN are iterative, while convolutional neural networks (CNNs) offer a learned approach to reconstruction [4, 5]. Related work has also investigated neural-network denoising of spectral cubes [6, 7]. These studies provide context for the present image-domain masked-completion task. In this project, both decoder refinement architecture and the effects of an auxiliary loss function comprising SSIM and positive-emission terms were examined.

2.2 Masked Autoencoders and Vision Transformers

Self-supervised reconstruction requires no manual labels, which is beneficial for radio cubes because annotation can be difficult and uncertain. Instead, a model learns a latent representation by reconstructing masked regions from the visible context. Pathak et al. [8] demonstrated representation learning through image inpainting. Masked autoencoders (MAEs) extend this approach by reconstructing missing image patches from an encoded visible subset [9]. Their Vision Transformer backbone can capture relationships beyond the immediate pixel neighbourhood [10], making this approach attractive for radio cubes. Kalkhof et al. [11] demonstrated the capabilities of neural cellular automata (NCA) for lightweight medical-image segmentation. Inspired by biological morphogenesis, NCA have also been used to generate and regenerate images through repeated local updates [12]. This motivates investigating an NCA refiner alongside the MAE's transformer-based representation.

2.3 Auxiliary Loss Function

Pointwise MSE measures voxel agreement but does not explicitly measure structural similarity. Under uncertainty, it can favour averaged predictions that smooth fine structure. SSIM provides a complementary measure of image structure [13]. Daw et al. [14] used physics-guided loss functions for lake temperature modelling, illustrating how scientific knowledge can inform an optimisation objective. Inspired by this broader idea, the present work added SSIM and a positive-emission term to emphasise structure and high-intensity target regions. Here, positive emission refers to target voxels above a noise-based threshold, rather than independently verified astronomical emission. These are lightweight empirical terms, rather than physical conservation constraints, and their scientific benefit must be evaluated rather than assumed.

Thus, this work examines the intersection of MAE decoder refinement and auxiliary loss design for masked radio-cube reconstruction.

3 Methodology

3.1 Task, data, and frozen split

This study addresses masked completion of already imaged and deconvolved PHANGS–ALMA CO(2–1) spectral cubes [2, 15]; it does not model visibilities, uv sampling, calibration, the synthesized beam, or deconvolution. The experiment is therefore an image-domain inpainting benchmark on science-ready cubes, not interferometric image reconstruction.

The frozen manifest contains 68 galaxy cubes: 44 training cubes, 19 validation cubes used only for checkpoint selection, and five independent test cubes (NGC 1672, NGC 2903, NGC 2997, NGC 3521, and NGC 4536). No test galaxy was used for training or checkpoint selection. The training set was expanded to 264 samples per epoch using the identity transform, rotations by 90°, 180°, and 270°, and horizontal and vertical reflections. The model input was the CO(2–1) cube; precomputed moment maps were not model inputs or targets.

3.2 Preprocessing and validity

FITS axes were identified from their CTYPE metadata and reordered to (ν, y, x). Singleton axes were removed. In the legacy preprocessing used by the frozen checkpoints, non-finite samples were replaced by zero before a per-cube mean and standard-deviation transform. Cubes were then cropped or zero-padded

to $64 \times 128 \times 128$ voxels. Training crops were signal-biased with probability 0.7; validation and test evaluation used deterministic central crops. The legacy validity tensor identifies geometrical padding, but does not preserve the original non-finite FITS footprint. This limitation applies to the normalized training losses and diagnostics. The calibrated analysis described below reloads the original FITS data and excludes non-finite voxels explicitly.

Each cube was divided into non-overlapping $4 \times 4 \times 4$ patches, giving a $16 \times 32 \times 32$ patch grid. All reported experiments hid exactly 50% of patches. Unless otherwise stated, the loss and normalized metrics were evaluated only on voxels belonging to hidden patches and marked valid by the legacy padding mask.

3.3 Shared masked-autoencoder backbone

All compared refiners used the same masked-autoencoder (MAE) backbone. A strided 3-D convolution projected each 64-voxel patch to a 192-dimensional token. Fixed three-dimensional sine–cosine positional embeddings were added before a four-block transformer encoder with eight attention heads. Only visible tokens entered the encoder. The encoded tokens were projected to 256 dimensions, combined with learned mask tokens in their original patch order, and processed by a four-block, eight-head transformer decoder. Layer normalization and a linear head produced 64 values per patch. The Linear reference stopped at this patch prediction. The Conv3D and NCA variants assembled a complete preliminary cube using true visible patches and MAE predictions in hidden patches, then applied their respective refinement module.

3.4 Refinement modules

3.4.1 Conv3D

The Conv3D refiner learned a residual correction to the preliminary cube. It comprised three $3 \times 3 \times 3$ convolutions with channel widths $1 \to 16 \to 16 \to 1$ and GELU activations after the first two convolutions. The final convolution was initialized to zero, so the initial module represented the identity mapping. This is a small local refiner, not a U-Net or a claim to represent all 3-D CNNs.

3.4.2 Neural cellular automaton

The implemented NCA used 16 hidden state channels and eight recurrent steps. At step t, its 20-channel voxel state was

$$\mathbf{S}^{(t)} = \left[\hat{I}^{(t)}, V, M, G, \mathbf{H}^{(t)}\right], \tag{1}$$

where $\hat{I}$ is the current intensity estimate, V and M are the visible- and hidden-patch indicator volumes, G is the legacy geometrical-validity mask, and $\mathbf{H}$ contains 16 mutable hidden channels. The initial intensity was the complete preliminary MAE reconstruction—true input values in visible patches and MAE predictions in hidden patches—and $\mathbf{H}^{(0)} = 0$. Thus the NCA did not begin from zeros in the hidden region.

Perception concatenated the state itself with two fixed depthwise responses:

$$\mathbf{P}^{(t)} = \left[\mathbf{S}^{(t)}, K_{xy} * \mathbf{S}^{(t)}, K_{\nu} * \mathbf{S}^{(t)}\right]. \tag{2}$$

K_{xy} was a five-point spatial Laplacian in the central spectral plane (centre -4, four spatial neighbours $+1$), and K_{ν} was a three-point spectral Laplacian (centre -2, two spectral neighbours $+1$). There were no learned perception kernels or first-derivative kernels. The resulting 60 channels passed through a shared $1 \times 1 \times 1$ transition network of width 32 with a GELU activation:

$$\mathbf{S}^{(t+1)} = \mathbf{S}^{(t)} + W_2\,\mathrm{GELU}(W_1\mathbf{P}^{(t)} + b_1) + b_2. \tag{3}$$

No additional update-scale parameter was used. The final transition layer was zero-initialized. After each of the eight updates, V, M, and G were reset to their fixed inputs and the intensity at visible voxels was clamped to the observed value. The output intensity was multiplied by G.

The complete models differed by 5,197 trainable parameters, approximately 0.10% of the Conv3D total, and were therefore closely parameter-matched. This does not imply compute matching because the NCA transition was applied recurrently for eight steps.

Table 1: Trainable parameter counts for the two refinement architectures within the shared MAE framework.

Architecture	Complete model	Refiner only
Conv3D	5,020,033	7,809
NCA	5,014,836	2,612

3.5 Training objectives

The baseline objective was mean squared error on the hidden valid voxels $\mathcal{M}$,

$$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{|\mathcal{M}|} \sum_{i \in \mathcal{M}} (\hat{y}_i - y_i)^2. \tag{4}$$

The auxiliary objective added slice-wise SSIM and positive-emission MSE. SSIM was computed independently in every spectral slice with a 7×7 spatial window; local statistics used hidden valid samples, the stability constants used the per-cube 1st–99th percentile target range, and populated slices were weighted equally. The positive-emission term selected hidden valid target values satisfying $y_i \geq 3\sigma_{\mathrm{MAD}}$, where $\sigma_{\mathrm{MAD}} = 1.4826\,\mathrm{median}(|y - \mathrm{median}(y)|)$ was estimated separately for each cube. If no value passed the threshold, this term was zero. The resulting auxiliary loss function was

$$\mathcal{L}_{\mathrm{aux}} = \mathcal{L}_{\mathrm{MSE}} + 0.05\mathcal{L}_{\mathrm{SSIM}} + 0.05\mathcal{L}_{\mathrm{emission}}. \tag{5}$$

It contains no governing equation or conservation constraint. Although the implementation can include a normalized channel-sum term, its weight was zero for every checkpoint reported here.

3.6 Training and checkpoint selection

The controlled Conv3D–NCA factorial used AdamW with learning rate 10^{-4}, weight decay 0.05, training seed 123, an effective batch size of four, and 40 epochs. With 264 augmented samples per epoch this gave 66 optimizer updates per epoch and 2640 updates in total. Conv3D used physical batch size four; the memory-intensive mixed-mask NCA runs used physical batch size two with two-step gradient accumulation. Random-only training used random patch masks. Mixed training sampled 70% random masks and 30% structured slab/block masks. For every architecture–loss–training-mask cell, the reported checkpoint was the epoch with the lowest signal-region MSE on the separate 19-cube validation set. Consequently, selected checkpoints range from step 2112 to step 2640 even though every run received the same total training budget.

3.7 Evaluation geometries and aggregation

The five held-out galaxies were evaluated under three 50% hidden-patch geometries. Random masks sampled half of all patches. Frequency-varying masks sampled half of the 32×32 spatial patches independently in each of the 16 spectral patch planes. Structured masks selected contiguous spectral slabs, sky-plane gaps, and a coupled spectral–spatial block. The structured geometry is a severe image-domain stress test, not a simulation of ALMA flagging or missing Fourier coverage.

Evaluation used seeds 123, 456, and 789. These are three deterministic mask realizations applied to the same trained checkpoint, not three training seeds or independently trained models. Metrics were first averaged across the three masks within each galaxy; reported means and sample standard deviations then use the five galaxies as the independent units.

3.8 Normalized and calibrated metrics

Normalized-space diagnostics included hidden-voxel MSE and MAE, PSNR, slice-wise SSIM, spectral-correlation MAE, and signal-region MSE/MAE. PSNR used the target maximum-minus-minimum range over valid voxels and the hidden-voxel MSE. Spectral-correlation MAE compared the predicted and target channel-correlation matrices on channel pairs with at least two jointly evaluated spatial samples and non-zero variance. The signal support was target-defined: valid target voxels at least three robust noise standard deviations from the median were retained only in channels containing at least eight such

voxels and integrated S/N at least 20. The normalized "moment-0 recovery" is the ratio of predicted to target channel sums on hidden, positive, target-defined signal voxels. It is a diagnostic in standardized units, not a calibrated moment or a molecular-gas measurement.

For the physical analysis, predictions were mapped back to kelvin by inverting each cube's affine training normalization. Original non-finite FITS voxels were excluded. The target-defined aperture comprised positive coherent emission from the same signal-support rule. Channel width was derived from the spectral WCS; integrated brightness was computed in $\mathrm{K\,km\,s^{-1}}$ and converted to $\mathrm{Jy\,km\,s^{-1}}$ with the Rayleigh–Jeans brightness-temperature factor at the FITS rest frequency and the celestial pixel solid angle. Hidden recovery is the predicted-to-target integrated flux ratio within masked aperture voxels. Full recovery replaces only masked voxels with predictions and retains observed emission exactly. Because the aperture is derived from the target, these values quantify reconstruction fidelity under a controlled benchmark; they do not demonstrate blind source finding, physical conservation, molecular-gas mass accuracy, or visibility-domain fidelity.

The implementation used Astropy, NumPy, and PyTorch [16–18]. Exact checkpoint hashes, the split-manifest hash, evaluation commands, mask seeds, and frozen output hashes are recorded in the submission provenance note.

4 Results

4.1 Controlled refiner comparison under random masking

Table 2: Optimizer-matched Conv3D–NCA comparison under 50% random masking on five independent test galaxies. Values are galaxy-level mean ± sample standard deviation after averaging the three deterministic evaluation-mask realizations per galaxy.

Model	Masked MSE	Masked MAE	Signal MSE	Signal MAE	PSNR (dB)	SSIM	Spectral-correlation MAE	Moment-0 recovery (%)
Conv3D	2.612 ± 1.473	0.676 ± 0.129	22.400 ± 11.657	2.841 ± 0.593	34.38 ± 1.78	0.835 ± 0.077	0.042 ± 0.015	92.91 ± 4.74
NCA	$\mathbf{1.560 \pm 1.002}$	$\mathbf{0.447 \pm 0.083}$	$\mathbf{13.760 \pm 8.749}$	$\mathbf{1.980 \pm 0.481}$	$\mathbf{36.80 \pm 1.65}$	$\mathbf{0.916 \pm 0.043}$	$\mathbf{0.030 \pm 0.014}$	$\mathbf{96.34 \pm 2.90}$

Under the controlled 40-epoch protocol, Conv3D obtained a signal-region MSE of 22.400, SSIM of 0.835, and normalized recovery of 92.91%. The corresponding NCA checkpoint obtained 13.760, 0.916, and 96.34%, respectively. These results observe an advantage for this NCA refiner over this small Conv3D refiner under the stated random-mask protocol; they do not establish superiority over arbitrary 3-D CNNs or U-Nets.

4.2 Auxiliary-loss ablation under random masking

Table 3: Controlled 40-epoch auxiliary-loss comparison under 50% random masking. Values are galaxy-level mean ± sample standard deviation after averaging three deterministic evaluation-mask realizations per galaxy.

Architecture	Loss function	Masked MSE	Masked MAE	Signal MSE	Signal MAE	PSNR (dB)	SSIM	Spectral-correlation MAE	Moment-0 recovery (%)
Conv3D	MSE	2.612 ± 1.473	0.676 ± 0.129	22.400 ± 11.657	2.841 ± 0.593	34.38 ± 1.78	0.835 ± 0.077	0.042 ± 0.015	92.91 ± 4.74
Conv3D	Auxiliary	2.481 ± 1.381	0.678 ± 0.124	20.981 ± 10.832	2.760 ± 0.583	34.58 ± 1.82	0.835 ± 0.079	0.043 ± 0.014	94.38 ± 4.10
NCA	MSE	1.560 ± 1.002	0.447 ± 0.083	13.760 ± 8.749	1.980 ± 0.481	36.80 ± 1.65	0.916 ± 0.043	0.030 ± 0.014	96.34 ± 2.90
NCA	Auxiliary	$\mathbf{1.244 \pm 0.796}$	$\mathbf{0.417 \pm 0.063}$	$\mathbf{10.705 \pm 7.089}$	$\mathbf{1.667 \pm 0.361}$	$\mathbf{37.71 \pm 1.82}$	$\mathbf{0.917 \pm 0.045}$	$\mathbf{0.028 \pm 0.011}$	95.82 ± 1.29

For Conv3D, the combined loss reduces signal MSE from 22.400 to 20.981 and changes moment-0 recovery from 92.91% to 94.38%. For the NCA, it reduces signal MSE from 13.760 to 10.705; SSIM changes from 0.916 to 0.917 and moment-0 recovery from 96.34% to 95.82%.

4.3 Mask distribution experiment

Table 4: Primary outcomes from the fair factorial comparison of random-only and mixed-mask (70% random, 30% structured slab/block) training. Values are galaxy-level mean ± sample standard deviation across five test galaxies after averaging three deterministic evaluation-mask seeds within each galaxy. The three seeds are mask realizations, not independent training runs.

Architecture	Training	Loss	Signal-region MSE			Moment-0 recovery (%)		
			Random	Freq.-varying	Structured	Random	Freq.-varying	Structured
Conv3D	Random only	MSE	22.400 ± 11.657	22.284 ± 12.786	130.695 ± 71.229	92.91 ± 4.74	93.06 ± 4.09	31.87 ± 3.87
Conv3D	Random only	Auxiliary	20.981 ± 10.832	20.963 ± 12.078	127.482 ± 69.245	94.38 ± 4.10	94.54 ± 3.45	33.28 ± 3.81
Conv3D	Mixed 70/30	MSE	27.955 ± 15.064	27.680 ± 16.342	$\mathbf{116.261 \pm 65.436}$	$\mathbf{99.38 \pm 4.73}$	$\mathbf{99.67 \pm 3.91}$	$\mathbf{41.90 \pm 2.19}$
Conv3D	Mixed 70/30	Auxiliary	24.253 ± 12.622	24.077 ± 13.895	119.767 ± 64.521	97.43 ± 4.65	97.69 ± 3.90	38.55 ± 3.49
NCA	Random only	MSE	13.760 ± 8.749	14.346 ± 9.156	157.769 ± 88.375	96.34 ± 2.90	96.51 ± 2.01	20.18 ± 5.74
NCA	Random only	Auxiliary	10.705 ± 7.089	11.174 ± 7.217	152.568 ± 86.613	95.82 ± 1.29	95.94 ± 0.51	22.96 ± 5.21
NCA	Mixed 70/30	MSE	12.372 ± 6.749	13.192 ± 8.399	131.398 ± 73.161	101.34 ± 3.23	101.56 ± 2.59	31.12 ± 4.80
NCA	Mixed 70/30	Auxiliary	$\mathbf{10.012 \pm 5.120}$	$\mathbf{10.611 \pm 6.498}$	119.345 ± 69.701	102.37 ± 2.26	102.55 ± 1.61	39.83 ± 2.81

Mixed training used 70% random masks and 30% structured slab/block masks. NCA with the auxiliary loss function and mixed training achieved the lowest signal MSE under random and frequency-varying evaluation, at 10.012 and 10.611, respectively. Under the harder structured evaluation, Conv3D with MSE and mixed training achieved the lowest signal MSE, 116.261, narrowly below the NCA auxiliary-loss mixed result of 119.345. Thus the NCA did not win every mask geometry. Complete secondary metrics are retained in the machine-readable final-factorial evaluation output.

4.4 Scientific information recovery

Table 5: Normalized-unit moment-0 diagnostic under the three evaluation masks. Values are galaxy-level mean ± sample standard deviation after averaging three deterministic evaluation-mask seeds per galaxy. The structured absolute error is $|\overline{R}_{\mathrm{flux}} - 100\%|$ and does not represent calibrated physical flux.

Model	Loss	Training	Random recovery (%)	Freq.-varying recovery (%)	Structured recovery (%)	Structured absolute error (%)	Structured moment-0 correlation
Conv3D	MSE	Random	92.91 ± 4.74	93.06 ± 4.09	31.87 ± 3.87	68.13	0.769 ± 0.252
Conv3D	Auxiliary	Random	94.38 ± 4.10	94.54 ± 3.45	33.28 ± 3.81	66.72	0.777 ± 0.255
Conv3D	MSE	Mixed 70/30	99.38 ± 4.73	99.67 ± 3.91	$\mathbf{41.90 \pm 2.19}$	**58.10**	0.798 ± 0.223
Conv3D	Auxiliary	Mixed 70/30	97.43 ± 4.65	97.69 ± 3.90	38.55 ± 3.49	61.45	0.787 ± 0.261
NCA	MSE	Random	96.34 ± 2.90	96.51 ± 2.01	20.18 ± 5.74	79.82	0.673 ± 0.183
NCA	Auxiliary	Random	95.82 ± 1.29	95.94 ± 0.51	22.96 ± 5.21	77.04	0.701 ± 0.185
NCA	MSE	Mixed 70/30	101.34 ± 3.23	101.56 ± 2.59	31.12 ± 4.80	68.88	0.776 ± 0.211
NCA	Auxiliary	Mixed 70/30	102.37 ± 2.26	102.55 ± 1.61	39.83 ± 2.81	60.17	$\mathbf{0.808 \pm 0.185}$

This normalized recovery ratio measures how closely the reconstructed hidden, target-defined signal sum matches the target sum. It complements local error and structural metrics, but it is computed after per-cube standardization and does not by itself measure column density, molecular-gas mass, or calibrated flux. In particular, the structured values are recovered fractions: the best normalized result is 41.90%, meaning that more than half of the hidden signal sum remains unrecovered.

4.5 Calibrated physical-unit evaluation

The final checkpoints were also evaluated after inverting each cube's training normalization. The FITS headers give $BUNIT = \mathrm{K}$, a channel width of approximately $2.5406\,\mathrm{km\,s^{-1}}$, the CO(2–1) rest frequency, and the celestial pixel scale. Integrated brightness was therefore calculated in $\mathrm{K\,km\,s^{-1}}$ and converted to $\mathrm{Jy\,km\,s^{-1}}$ using the Rayleigh–Jeans brightness-temperature conversion and each cube's pixel solid angle. The aperture was defined from positive coherent target emission and excluded non-finite FITS voxels. Three deterministic mask realizations were averaged within each galaxy before calculating the five-galaxy mean and sample standard deviation.

Table 6: Calibrated integrated-flux recovery. Hidden recovery evaluates only masked target-defined emission; full structured recovery includes observed emission retained exactly outside the mask. Absolute error is calculated in the hidden structured aperture. Values are five-galaxy mean ± sample standard deviation after averaging three evaluation-mask seeds per galaxy.

Model	Loss	Training	Random hidden recovery (%)	Freq.-varying hidden recovery (%)	Structured hidden recovery (%)	Structured full recovery (%)	Structured hidden absolute error ($\mathrm{Jy\,km\,s^{-1}}$)
Conv3D	MSE	Random	92.94 ± 4.70	93.09 ± 4.05	32.14 ± 3.75	66.22 ± 3.94	355.06 ± 165.96
Conv3D	Auxiliary	Random	94.40 ± 4.07	94.56 ± 3.42	33.54 ± 3.68	66.92 ± 3.86	347.79 ± 162.82
Conv3D	MSE	Mixed 70/30	$\mathbf{99.39 \pm 4.71}$	$\mathbf{99.68 \pm 3.89}$	$\mathbf{42.13 \pm 2.17}$	$\mathbf{71.14 \pm 3.72}$	$\mathbf{299.93 \pm 127.77}$
Conv3D	Auxiliary	Mixed 70/30	97.45 ± 4.62	97.71 ± 3.88	38.79 ± 3.38	69.49 ± 4.10	319.54 ± 146.31
NCA	MSE	Random	96.36 ± 2.88	96.52 ± 1.99	20.50 ± 5.60	60.52 ± 3.77	417.27 ± 202.16
NCA	Auxiliary	Random	95.84 ± 1.29	95.96 ± 0.51	23.26 ± 5.09	61.90 ± 3.42	402.12 ± 191.90
NCA	MSE	Mixed 70/30	101.34 ± 3.21	101.56 ± 2.58	31.39 ± 4.68	65.89 ± 3.70	359.49 ± 172.17
NCA	Auxiliary	Mixed 70/30	102.36 ± 2.26	102.55 ± 1.61	40.06 ± 2.82	70.20 ± 2.89	310.80 ± 134.01

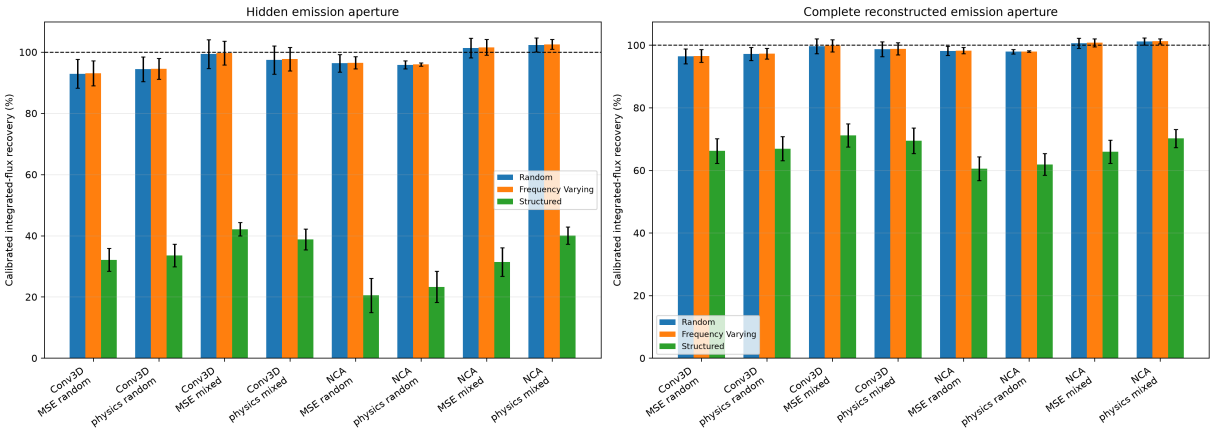


Figure 1: Calibrated integrated-flux recovery on the hidden emission aperture and the complete reconstructed emission aperture. Error bars are the sample standard deviation across five galaxies after averaging three deterministic evaluation-mask seeds per galaxy.

The calibrated analysis reproduces the main normalized-space ordering but also exposes the severity of contiguous missing regions. Conv3D with MSE and mixed training is closest to complete recovery for random and frequency-varying masks and gives the best structured hidden recovery, 42.13%. NCA with the auxiliary loss function and mixed training is second under structured masking at 40.06%. Thus none of the models provides reliable calibrated flux recovery for the structured stress test.

4.6 Per galaxy statistics

Table 7: Per-galaxy fair comparison of random-trained Conv3D and NCA under random masking, after averaging the three deterministic evaluation-mask seeds. Differences are NCA minus Conv3D; flux differences are percentage points.

Galaxy	Loss	Signal MSE			Normalized moment-0 recovery (%)		
		Conv3D	NCA	Difference	Conv3D	NCA	Difference
NGC4536	MSE	38.001	27.461	-10.540	95.01	95.91	0.90
NGC3521	MSE	12.435	6.823	-5.612	89.34	93.52	4.18
NGC2997	MSE	29.016	15.836	-13.180	96.60	99.20	2.60
NGC2903	MSE	22.681	13.011	-9.670	97.11	99.47	2.36
NGC1672	MSE	9.869	5.669	-4.200	86.47	93.60	7.13
NGC4536	Auxiliary	35.201	22.357	-12.844	96.49	94.54	-1.95
NGC3521	Auxiliary	11.640	5.455	-6.185	91.82	94.68	2.86
NGC2997	Auxiliary	27.592	11.563	-16.029	97.32	96.57	-0.75
NGC2903	Auxiliary	21.159	9.374	-11.785	97.84	97.61	-0.23
NGC1672	Auxiliary	9.315	4.776	-4.539	88.41	95.73	7.32

The five independent test galaxies are treated as independent units. Galaxy-level paired comparisons were used because voxel-level observations within a cube are dependent. Under random masking, the NCA reduced signal MSE for every galaxy under both losses. Normalized moment-0 recovery was closer to 100% in all five galaxies for MSE and two of five galaxies for the auxiliary loss function. The two-sided

Table 8: Summary of paired NCA-minus-Conv3D differences for random-trained models under random masking. Results use five independent galaxies after averaging three deterministic evaluation-mask seeds. Wilcoxon tests are exact and two-sided.

Loss	Metric	Mean ± SD	Median	Range	W	p
MSE	Signal MSE	-8.640 ± 3.680	-9.670	$[-13.180, -4.200]$	0	0.0625
MSE	Recovery (pp)	3.435 ± 2.370	2.598	$[0.903, 7.129]$	0	0.0625
Auxiliary	Signal MSE	-10.276 ± 4.786	-11.785	$[-16.029, -4.539]$	0	0.0625
Auxiliary	Recovery (pp)	1.450 ± 3.729	-0.231	$[-1.950, 7.315]$	6	0.8125

Wilcoxon tests returned $p = 0.0625$ for both signal-MSE comparisons. With only five non-zero pairs, 0.0625 is the minimum attainable exact two-sided value, so the conventional $p < 0.05$ threshold cannot be crossed.

4.7 Qualitative Results

NGC 4536 was fixed as the representative visualization independently of model performance. The zero-indexed channels 0, 15, 31, 47, and 63 span the 64-slice crop. Within each comparison, all models receive the same deterministic hidden patches; the frequency-varying geometry changes the spatial pattern between spectral patch planes. These panels illustrate one galaxy and must be interpreted alongside the five-galaxy aggregate results.

4.7.1 Random Masking

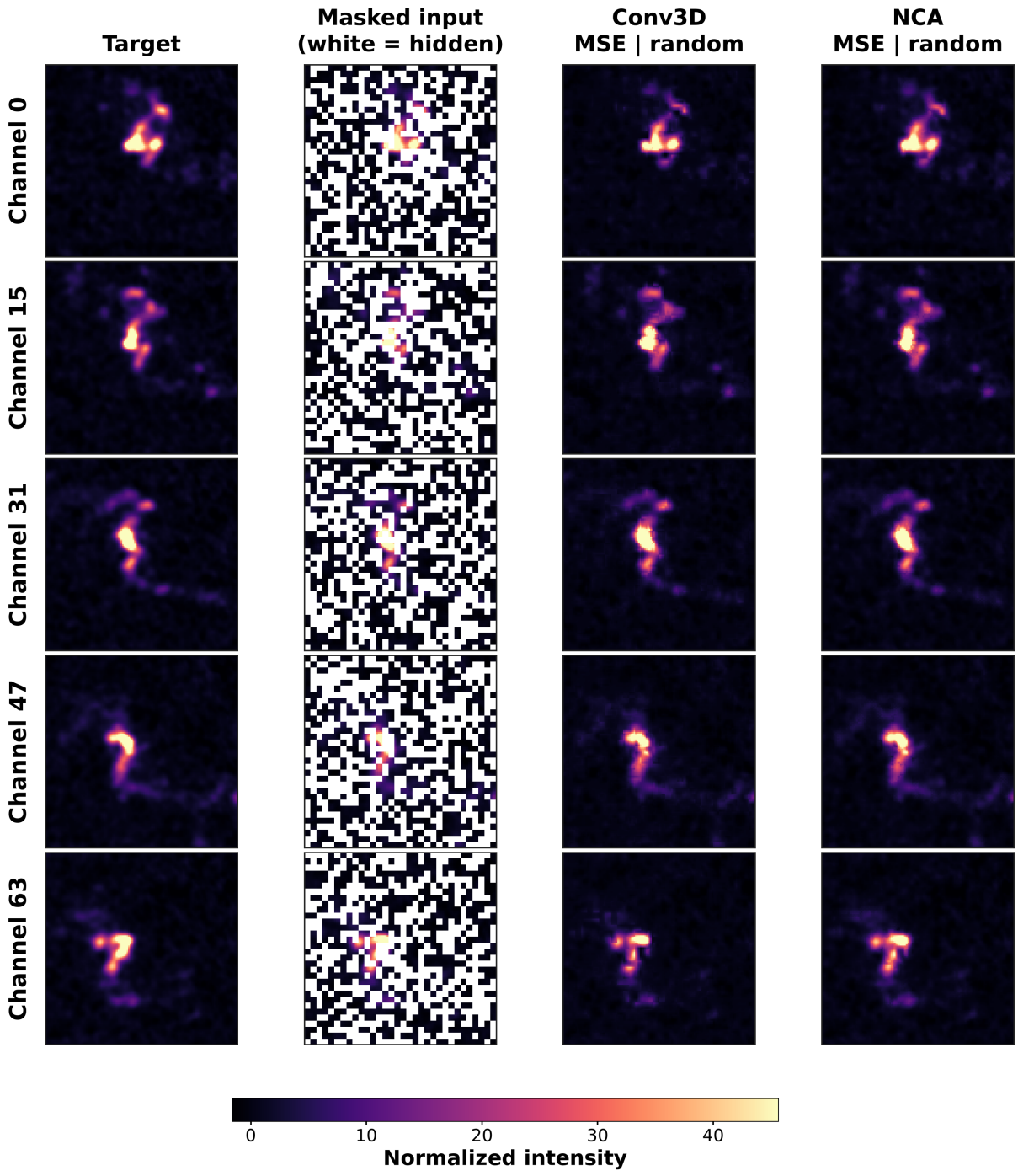


Figure 2: Random-mask comparison for the fair MSE-decoder models. Columns follow the conventional target–input–reconstruction order, so each bold model heading is directly above its output. White pixels in the masked-input column are hidden; all panels use one shared robust intensity scale.

Under the displayed random mask, both refiners recover the location of the principal emission, while the NCA panel is smoother and has smaller visible residuals around the bright structure. A single target-derived display cannot establish the absence of hallucinations; plausible small artifacts can evade visual inspection [19]. The qualitative ordering is consistent with the lower aggregate random-mask MSE and higher SSIM of the NCA checkpoint.

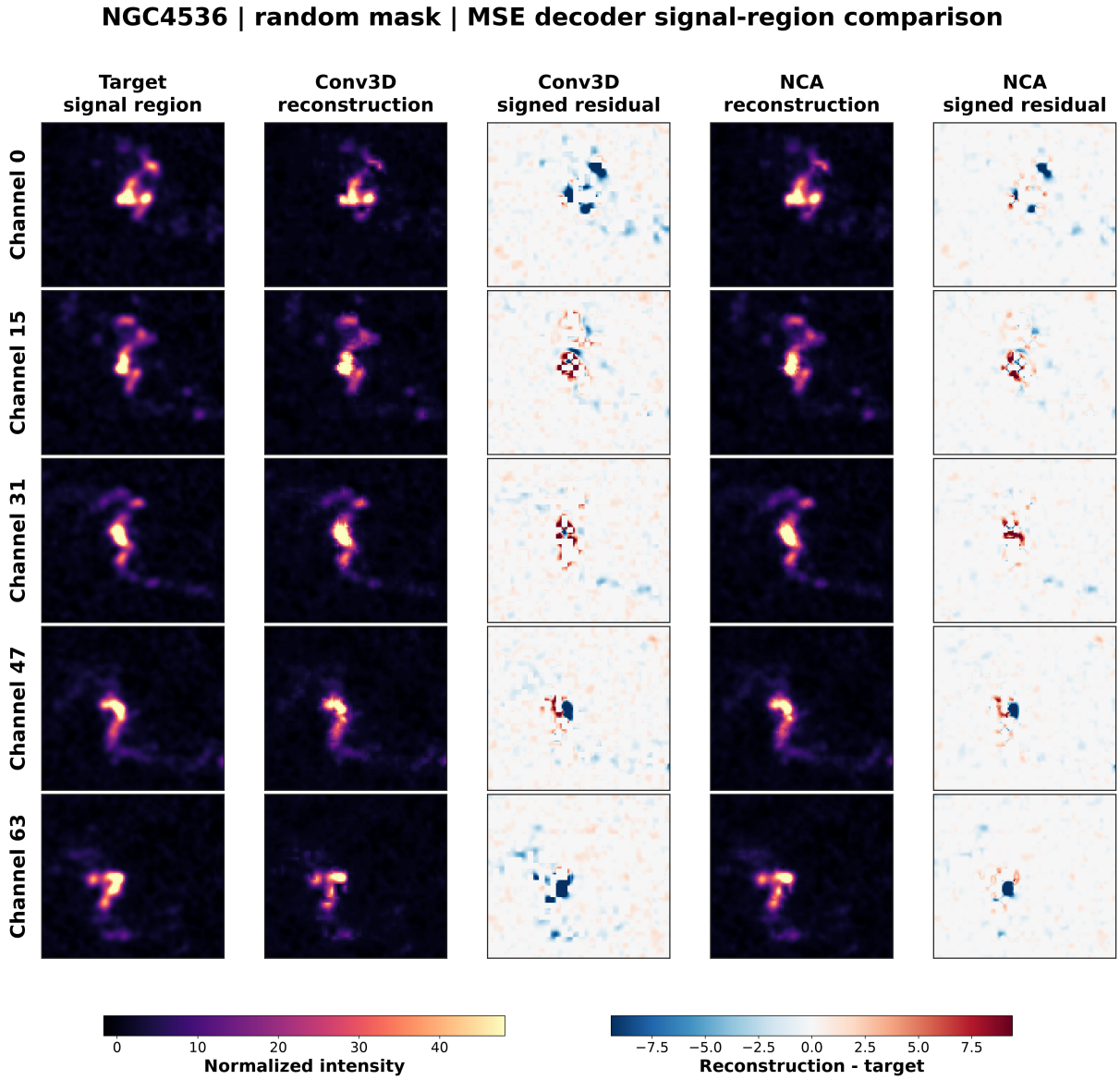


Figure 3: Signal-region reconstructions and signed residuals for the same fair MSE-decoder comparison. Each reconstruction is immediately followed by its residual (reconstruction minus target). Reconstruction panels share one intensity scale and residual panels share one symmetric scale.

Residuals are defined as reconstruction minus target, so negative values indicate under-recovery. In this example the largest residuals occur near bright target emission, and the Conv3D errors cover a larger visible area than the NCA errors. This is descriptive evidence for NGC 4536 under one mask realization, not an independent statistical result.

Channel 63 is reconstructed poorly by both displayed checkpoints, but the panel alone does not identify the cause; the hidden support and boundary effects are confounded. No claim of artifact-free reconstruction is made.

4.7.2 Frequency-varying masking

The frequency-varying generator hides exactly half of the spatial patches independently within each spectral patch plane. It therefore preserves contextual coverage across the cube and is distinct from the contiguous structured slab/block stress test. In the displayed example, changing from random-only to mixed training produces only subtle visual changes for either architecture. The aggregate signal-region MSE, rather than this example, provides the quantitative comparison.

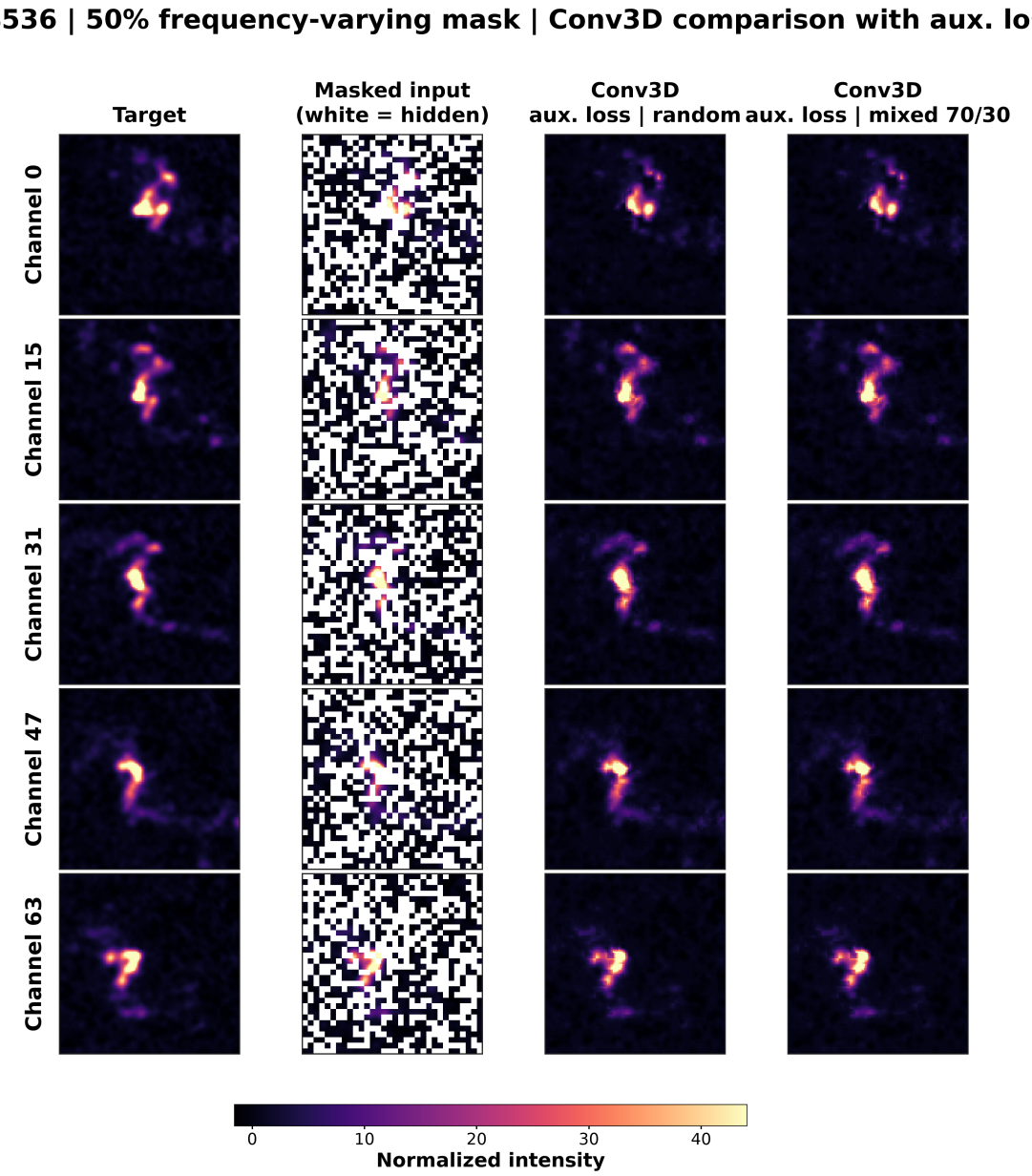


Figure 4: Frequency-varying-mask comparison for the auxiliary-loss Conv3D models trained with random-only or mixed 70/30 masking. The target and masked input appear first, followed immediately by the two labelled reconstructions; white denotes hidden input and all panels share one robust intensity scale.

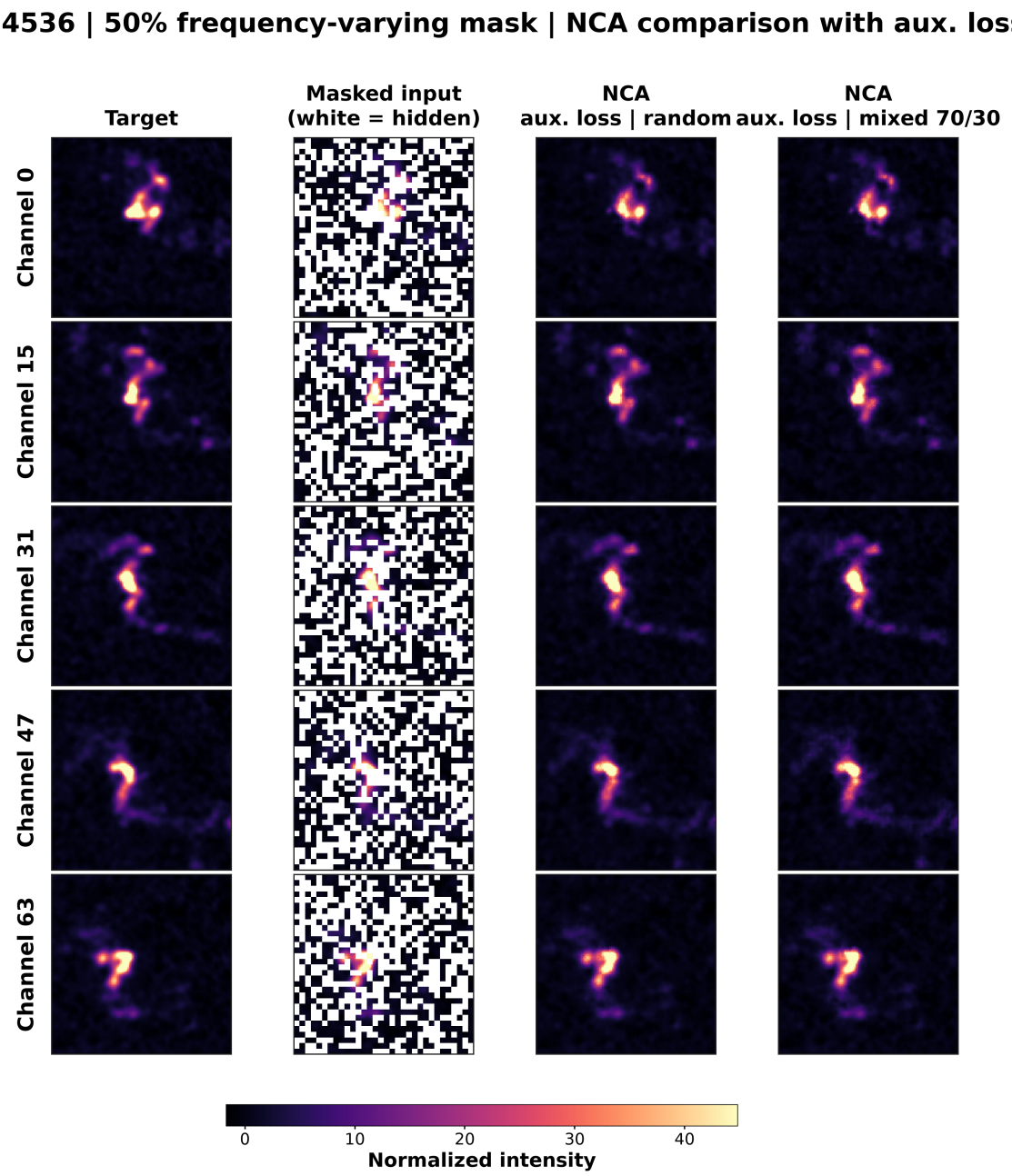


Figure 5: Frequency-varying-mask comparison for the auxiliary-loss NCA models trained with random-only or mixed 70/30 masking, using the same column order, evaluation mask, and channels as Figure 4. All panels within this figure share one robust intensity scale.

4.7.3 Normalized channel-sum reconstruction and residual analysis

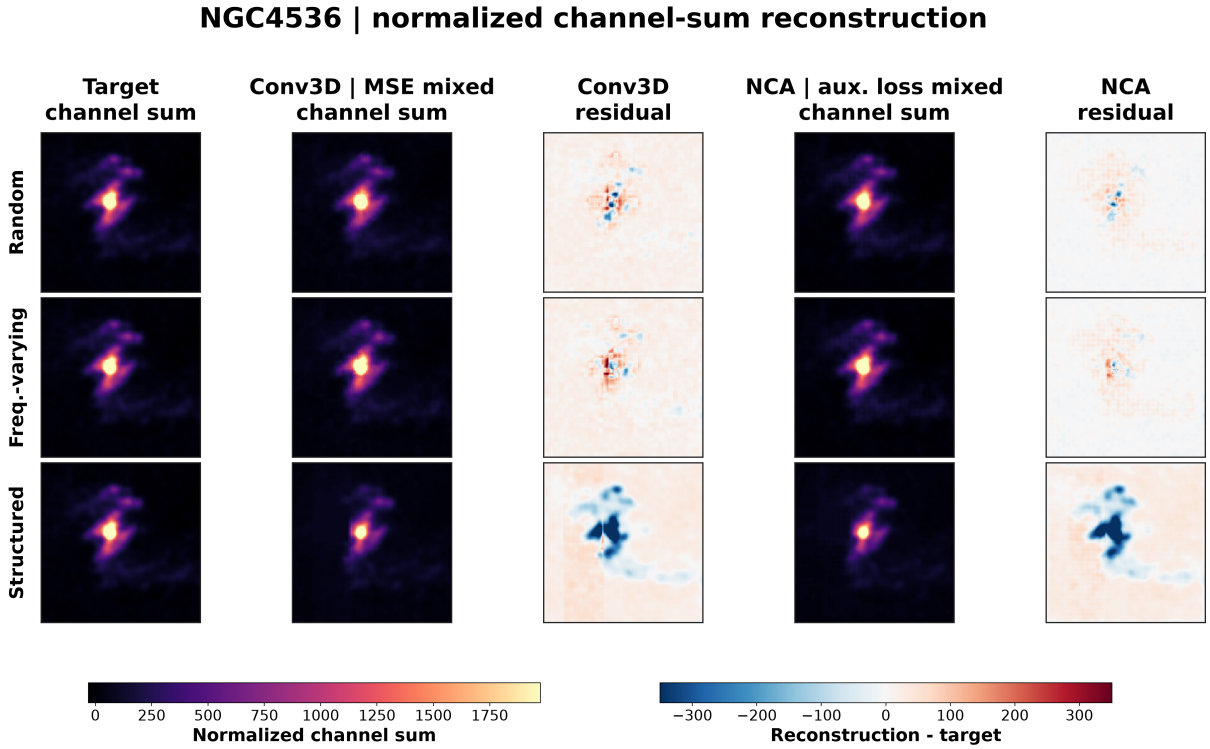


Figure 6: Normalized channel-sum reconstructions and signed residuals for Conv3D MSE mixed training and NCA auxiliary-loss mixed training under all three evaluation masks. Intensity and residual columns use separate shared scales. These are normalized diagnostic sums rather than calibrated physical fluxes.

The normalized channel-sum residuals show that errors concentrate near the brightest target emission. Their sign and spatial distribution vary with architecture, training mask, and evaluation geometry, so this single example should not be used to rank the models. The aggregate tables provide the ranking: NCA with the auxiliary loss and mixed training gives the lowest signal-region MSE for random and frequency-varying masks, whereas Conv3D MSE with mixed training gives the strongest structured-mask recovery. All configurations fail to recover more than half of the hidden calibrated flux in the structured stress test.

4.8 Signed residual-bias analysis

Integrated-flux recovery is determined by the net signed residual and can therefore appear accurate when positive and negative voxel residuals cancel. To test whether the Conv3D MSE mixed model's near-100% recovery under random and frequency-varying masking reflected genuinely smaller errors, residuals were decomposed within the hidden target-defined emission aperture. For residual $r_i = \hat{y}_i - y_i$, the positive and negative components were summed separately. Their absolute sum measures gross residual flux, the signed sum measures net bias, and the cancellation fraction was defined as $1 - |\sum_i r_i| / \sum_i |r_i|$. All quantities in Table 9 are normalized by the target flux in the same aperture, except the dimensionless cancellation fraction.

Conv3D's net bias was closer to zero under both masks, consistent with its recovery ratio being nearer 100%. However, this did not reflect smaller residuals: its gross absolute residual was approximately 29% of target flux, compared with approximately 17% for NCA auxiliary mixed. The Conv3D positive and negative components were both about 14–15% and cancelled by 87–88%, whereas NCA showed a smaller gross residual with a modest net overestimate of 2.4–2.5%. Thus Conv3D's apparent advantage in integrated-flux recovery under these two masks is primarily a cancellation effect; NCA auxiliary mixed has the lower residual error by this measure.

Table 9: Signed residual-flux decomposition for the two headline mixed-training models, excluding the structured mask. Values are galaxy-level mean ± sample standard deviation after averaging the three deterministic evaluation-mask realizations within each of the five test galaxies. Positive and negative residual columns are reported as non-negative magnitudes relative to target flux.

Model	Evaluation mask	Positive residual (%)	Negative residual (%)	Gross absolute residual (%)	Net bias (%)	Cancellation (%)
Conv3D MSE mixed	Random	14.16 ± 1.62	14.78 ± 3.39	28.94 ± 2.46	-0.61 ± 4.71	87.17 ± 7.21
NCA auxiliary mixed	Random	9.58 ± 0.79	7.22 ± 1.71	16.80 ± 1.40	2.36 ± 2.26	80.36 ± 8.79
Conv3D MSE mixed	Frequency varying	14.28 ± 1.48	14.60 ± 2.94	28.87 ± 2.55	-0.32 ± 3.89	87.92 ± 4.01
NCA auxiliary mixed	Frequency varying	9.64 ± 0.77	7.09 ± 1.18	16.74 ± 1.17	2.55 ± 1.61	83.87 ± 9.18

4.9 Test-set scope

All headline metrics use the five independent test galaxies, none of which was used for training or checkpoint selection. Conv3D and NCA each received the same 40-epoch, 2640-update budget; selection by validation signal-region MSE produced different retained epochs across factorial cells. The three evaluation seeds vary only the deterministic hidden-patch realization. They do not measure training-seed variability, and the five-galaxy sample does not support population-level generalization.

5 Discussion

The principal finding is conditional on the tested implementations and masks. With a shared MAE backbone, matched effective batch size, and matched optimizer-update budget, the NCA refiner obtained lower signal-region error than the small Conv3D refiner under random and frequency-varying masking. Under the severe structured slab/block stress test, the ordering changed: Conv3D MSE with mixed training gave the lowest signal-region MSE and the highest calibrated hidden-flux recovery. The experiment therefore does not demonstrate universal NCA superiority.

The metric suite separates several aspects of reconstruction. MSE and MAE quantify voxel error, the target-defined signal metrics focus attention on coherent high-amplitude emission, SSIM measures slice-wise morphology, and spectral-correlation MAE tests channel-to-channel relationships. The calibrated analysis then asks whether integrated emission is retained after restoring FITS units. Agreement across these diagnostics supports the random- and frequency-varying-mask result, while the structured-mask flux deficit exposes a clear failure mode. The data are consistent with iterative local refinement being useful in this particular parameter-matched comparison, but they do not isolate recurrence as the cause: the NCA and Conv3D also differ in effective depth and receptive field, and no compute-matched U-Net or deeper Conv3D was tested.

The auxiliary SSIM/emission objective reduced signal-region MSE for both random-trained architectures, although the size and direction of its effect depended on the metric and mask. It should therefore be described as an empirical loss ablation, not a physical constraint. Overall, the NCA advantage is observed only for the random and frequency-varying protocols. Robustness to the structured geometry is not supported, and higher-noise robustness was not tested.

5.1 Scientific Fidelity of Reconstruction

The moment-0 map,

$$M_0(x, y) = \sum_v I(x, y, v)\, \Delta v,$$

gives the velocity-integrated line intensity and traces the spatial distribution of integrated CO emission. Small errors across multiple channels can accumulate in M_0, even when the voxel-level MSE is low. Moment-0 recovery therefore measures whether reconstructed cubes retain integrated emission within the specified aperture. The controlled Conv3D and NCA results show that low voxel error and accurate integrated recovery need not rank configurations identically. The auxiliary SSIM/emission loss sometimes reduced signal-region error, but it did not consistently move calibrated recovery closer to 100% across architectures and masks. It therefore provides an empirical training preference rather than evidence of physical conservation. Nevertheless, moment-0 preservation alone does not guarantee physical fidelity. A model could recover the correct integrated flux while redistributing emission incorrectly between velocity channels. Consequently, moment-0 recovery should be interpreted alongside other information

such as channel residuals, spectral-correlation error and when useful the line-profile measurements. The normalized-unit diagnostics in Table 5 should not be interpreted as calibrated fluxes. Table 6 separately restores the FITS brightness-temperature scale and reports integrated flux in $\mathrm{Jy\,km\,s^{-1}}$; it still does not imply a molecular-gas mass without an excitation model and CO-to-H_2 conversion factor.

5.2 Limitations

5.2.1 Methodological

The dataset contains 68 galaxies but only five independent test galaxies, all from the same survey and CO transition. This is insufficient to establish population-level or cross-survey generalization. Although these galaxies were excluded from training and checkpoint selection, the same held-out set was inspected in development-stage comparisons; the results are therefore exploratory rather than a prospectively blinded confirmation. Only one training seed and one frozen galaxy split were used; seeds 123, 456, and 789 vary evaluation masks rather than learned weights. With five non-zero paired differences, the minimum attainable exact two-sided Wilcoxon value is $p = 0.0625$, so effect direction and size are more informative here than thresholded significance.

Training uses legacy per-cube standardized inputs in which non-finite samples were zero-filled before normalization and were not excluded by the training validity tensor. The physical evaluation corrects the aperture validity but cannot retroactively remove this influence from training. Full-cube normalization also uses values that are subsequently hidden by the synthetic mask; estimating normalization statistics for genuinely missing observations would require a separate procedure. The target-derived emission aperture is an oracle support, not a blind detection procedure. Post-hoc calibration tests integrated intensity but neither enforces conservation nor supplies an interferometric measurement or noise model. The implementation includes a Linear patch head, but the controlled result tables compare only the small Conv3D and NCA refiners; interpolation, U-Net, and depth/receptive-field/compute-matched convolutional controls are absent.

5.2.2 Computational

The NVIDIA RTX 4080 SUPER's 16 GB of GPU memory limited batch and cube sizes. Training time constrained the extent of hyperparameter optimisation. Storage was another limitation because individual downloaded FITS cubes could approach 1 GB.

5.3 Future Work

Further work could expand the dataset to another survey and use larger cubes and effective batch sizes where computational resources permit. Future galaxy-level splits could be stratified by galaxy properties; this was not ensured in the present study. A systematic hyperparameter study could examine learning rate, mask ratio, encoder dimension, and training duration for both architectures. Another extension is to characterise the relationship between conventional reconstruction error and error in scientific functionals. This would shift evaluation from visual and voxel-level fidelity towards scientific inference. Such analysis should also use a realistic interferometric measurement model, where measurement ambiguity may permit multiple reconstructions with different scientific interpretations.

6 Conclusion

In conclusion, the NCA refiner improved masked spatial reconstruction of imaged PHANGS–ALMA CO(2–1) cubes under random and frequency-varying masking relative to the Conv3D refiner within the matched training protocol. Under the harder structured masking condition, however, the best Conv3D configuration achieved slightly lower signal-region error, although neither architecture reconstructed the missing emission accurately. More broadly, the results demonstrate that stronger voxel-level reconstruction does not necessarily correspond to better scientific recovery of astronomical emission. Given the five-galaxy test set and the minimum attainable exact two-sided Wilcoxon p-value of 0.0625, these findings should be interpreted as directional rather than statistically conclusive.

Acknowledgements

This paper makes use of the following ALMA data: `ADS/JAO.ALMA#2012.1.00650.S`, `ADS/JAO.ALMA#2015.1.00925.S`, `ADS/JAO.ALMA#2015.1.00956.S`, `ADS/JAO.ALMA#2017.1.00886.L`, `ADS/JAO.ALMA#2018.1.01651.S`. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, OpenAI Codex was used to assist with drafting and debugging analysis code, code review, consistency checks, and revision of manuscript language. All code and text was reviewed and edited independently by the author. The author takes full responsibility for the accuracy, originality, and integrity of the work.

References

[1] Alberto D. Bolatto, Mark Wolfire, and Adam K. Leroy. The CO-to-H2 conversion factor. *Annual Review of Astronomy and Astrophysics*, 51(1):207–268, 2013. doi: 10.1146/annurev-astro-082812-140944.

[2] Adam K. Leroy, Eva Schinnerer, Annie Hughes, Erik Rosolowsky, Jérôme Pety, Andreas Schruba, Antonio Usero, Guillermo A. Blanc, Mélanie Chevance, Eric Emsellem, Christopher M. Faesi, Cinthya N. Herrera, Daizhong Liu, Sharon E. Meidt, Miguel Querejeta, Toshiki Saito, Karin M. Sandstrom, Jiayi Sun, Thomas G. Williams, Gagandeep S. Anand, Ashley T. Barnes, Erica A. Behrens, Francesco Belfiore, Samantha M. Benincasa, Ivana Bešlić, Frank Bigiel, Alberto D. Bolatto, Jakob S. den Brok, Yixian Cao, Rupali Chandar, Jérémy Chastenet, I-Da Chiang, Enrico Congiu, Daniel A. Dale, Sinan Deger, Cosima Eibensteiner, Oleg V. Egorov, Axel García-Rodríguez, Simon C. O. Glover, Kathryn Grasha, Jonathan D. Henshaw, I-Ting Ho, Amanda A. Kepley, Jaeyeon Kim, Ralf S. Klessen, Kathryn Kreckel, Eric W. Koch, J. M. Diederik Kruijssen, Kirsten L. Larson, Janice C. Lee, Laura A. Lopez, Josh Machado, Ness Mayker, Rebecca McElroy, Eric J. Murphy, Eve C. Ostriker, Hsi-An Pan, Ismael Pessa, Johannes Puschnig, Alessandro Razza, Patricia Sánchez-Blázquez, Francesco Santoro, Amy Sardone, Fabian Scheuermann, Kazimierz Sliwa, Mattia C. Sormani, Sophia K. Stuber, David A. Thilker, Jordan A. Turner, Dyas Utomo, Elizabeth J. Watkins, and Bradley Whitmore. PHANGS–ALMA: Arcsecond CO(2–1) imaging of nearby star-forming galaxies. *The Astrophysical Journal Supplement Series*, 257(2):43, 2021. doi: 10.3847/1538-4365/ac17f3.

[3] Jiayi Sun, Adam K. Leroy, Eva Schinnerer, Annie Hughes, Erik Rosolowsky, Miguel Querejeta, Andreas Schruba, Daizhong Liu, Toshiki Saito, Cinthya N. Herrera, Christopher Faesi, Antonio Usero, Jérôme Pety, J. M. Diederik Kruijssen, Eve C. Ostriker, Frank Bigiel, Guillermo A. Blanc, Alberto D. Bolatto, Médéric Boquien, Mélanie Chevance, Daniel A. Dale, Sinan Deger, Eric Emsellem, Simon C. O. Glover, Kathryn Grasha, Brent Groves, Jonathan Henshaw, Maria J. Jimenez-Donaire, Jenny J. Kim, Ralf S. Klessen, Kathryn Kreckel, Janice C. Lee, Sharon Meidt, Karin Sandstrom, Amy E. Sardone, Dyas Utomo, and Thomas G. Williams. Molecular gas properties on cloud scales across the local star-forming galaxy population. *The Astrophysical Journal Letters*, 901(1):L8, 2020. doi: 10.3847/2041-8213/abb3be.

[4] Kevin Schmidt, Felix Geyer, Stefan Fröse, Paul-Simon Blomenkamp, Marcus Brüggen, Francesco de Gasperin, Dominik Elsässer, and Wolfgang Rhode. Deep learning-based imaging in radio interferometry. *Astronomy and Astrophysics*, 664:A134, 2022. doi: 10.1051/0004-6361/202142113.

[5] Jan A. Högbom. Aperture synthesis with a non-regular distribution of interferometer baselines. *Astronomy and Astrophysics Supplement Series*, 15:417–426, 1974.

[6] Lucas Einig, Jérôme Pety, Antoine Roueff, Paul Vandame, Jocelyn Chanussot, Maryvonne Gerin, Jan H. Orkisz, Pierre Palud, Miriam G. Santa-Maria, Victor de Souza Magalhaes, Ivana Bešlić, Sébastien Bardeau, Emeric Bron, Pierre Chainais, Javier R. Goicoechea, Pierre Gratier, Viviana V. Guzmán, Annie Hughes, Jouni Kainulainen, David Languignon, Rosine Lallement, François Levrier, Dariusz C. Lis, Harvey S. Liszt, Jacques Le Bourlot, Franck Le Petit, Karin Öberg, Nicolas Peretto, Evelyne Roueff, Albrecht Sievers, Pierre-Antoine Thouvenin, and Pascal Tremblin. Deep learning denoising by dimension reduction: Application to the ORION-B line cubes. *Astronomy and Astrophysics*, 677:A158, 2023. doi: 10.1051/0004-6361/202346064.

[7] Arnab Lahiry, Tanio Díaz-Santos, Jean-Luc Starck, Niranjan Chandra Roy, Daniel Anglés-Alcázar, Grigorios Tsagkatakis, and Panagiotis Tsakalides. Deep and sparse denoising benchmarks for spectral data cubes of high-z galaxies: From simulations to ALMA observations, 2026. Preprint.

[8] Deepak Pathak, Philipp Krähenbühl, Jeff Donahue, Trevor Darrell, and Alexei A. Efros. Context encoders: Feature learning by inpainting. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 2536–2544, 2016. doi: 10.1109/CVPR.2016.278.

[9] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. Masked autoencoders are scalable vision learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 15979–15988, 2022. doi: 10.1109/CVPR52688.2022.01553.

[10] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=YicbFdNTTy.

[11] John Kalkhof, Camila González, and Anirban Mukhopadhyay. Med-NCA: Robust and lightweight segmentation with neural cellular automata. In *Information Processing in Medical Imaging*, volume 13939 of *Lecture Notes in Computer Science*, pages 705–716. Springer, 2023. doi: 10.1007/978-3-031-34048-2_54.

[12] Alexander Mordvintsev, Ettore Randazzo, Eyvind Niklasson, and Michael Levin. Growing neural cellular automata. *Distill*, 5(2), 2020. doi: 10.23915/distill.00023. URL https://distill.pub/2020/growing-ca/.

[13] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. doi: 10.1109/TIP.2003.819861.

[14] Arka Daw, Anuj Karpatne, William Watkins, Jordan Read, and Vipin Kumar. Physics-guided neural networks (PGNN): An application in lake temperature modeling, 2017.

[15] Adam K. Leroy, Annie Hughes, Daizhong Liu, Jérôme Pety, Erik Rosolowsky, Toshiki Saito, Eva Schinnerer, Andreas Schruba, Antonio Usero, Christopher M. Faesi, Cinthya N. Herrera, Mélanie Chevance, Alexander P. S. Hygate, Amanda A. Kepley, Eric W. Koch, Miguel Querejeta, Kazimierz Sliwa, David Will, Christine D. Wilson, et al. PHANGS–ALMA data processing and pipeline. *The Astrophysical Journal Supplement Series*, 255(1):19, 2021. doi: 10.3847/1538-4365/abec80.

[16] Astropy Collaboration, Adrian M. Price-Whelan, Pey Lian Lim, Nicholas Earl, Nathaniel Starkman, Larry Bradley, David L. Shupe, Aarya A. Patil, Lia Corrales, C. E. Brasseur, et al. The Astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package. *The Astrophysical Journal*, 935(2):167, 2022. doi: 10.3847/1538-4357/ac7c74.

[17] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020. doi: 10.1038/s41586-020-2649-2.

[18] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zachary DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An imperative style, high-performance deep learning library. In *Advances in Neural Information Processing Systems*, volume 32, pages 8024–8035, 2019. URL https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html.

[19] Nina M. Gottschling, Vegard Antun, Anders C. Hansen, and Ben Adcock. The troublesome kernel: On hallucinations, no free lunches, and the accuracy–stability tradeoff in inverse problems. *SIAM Review*, 67(1):73–104, 2025. doi: 10.1137/23M1568739.

A Additional evaluation figures



Figure 7: The three deterministic 50% hidden-patch evaluation geometries used in the final factorial comparison.

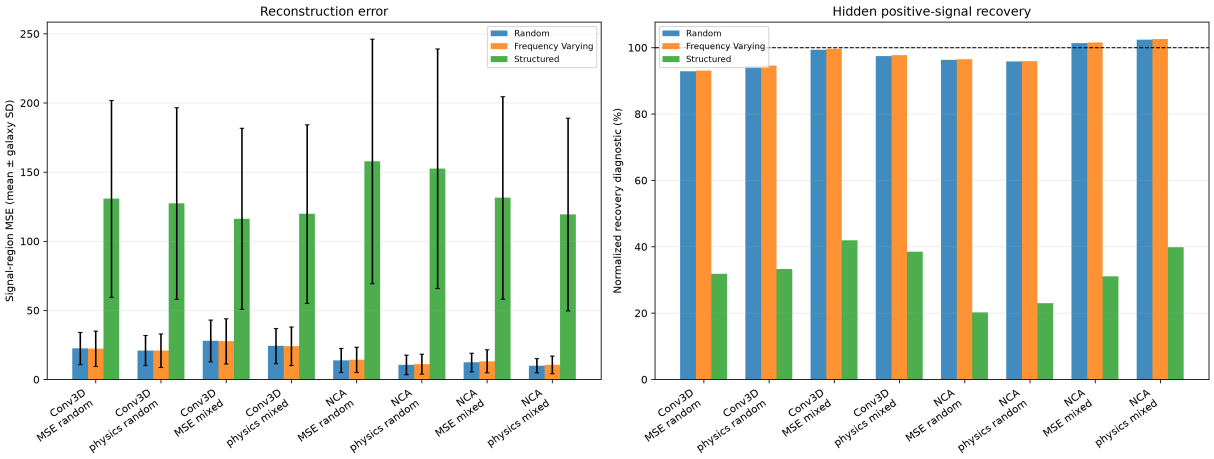


Figure 8: Signal-region MSE and normalized-unit recovery across all final factorial checkpoints and evaluation masks.

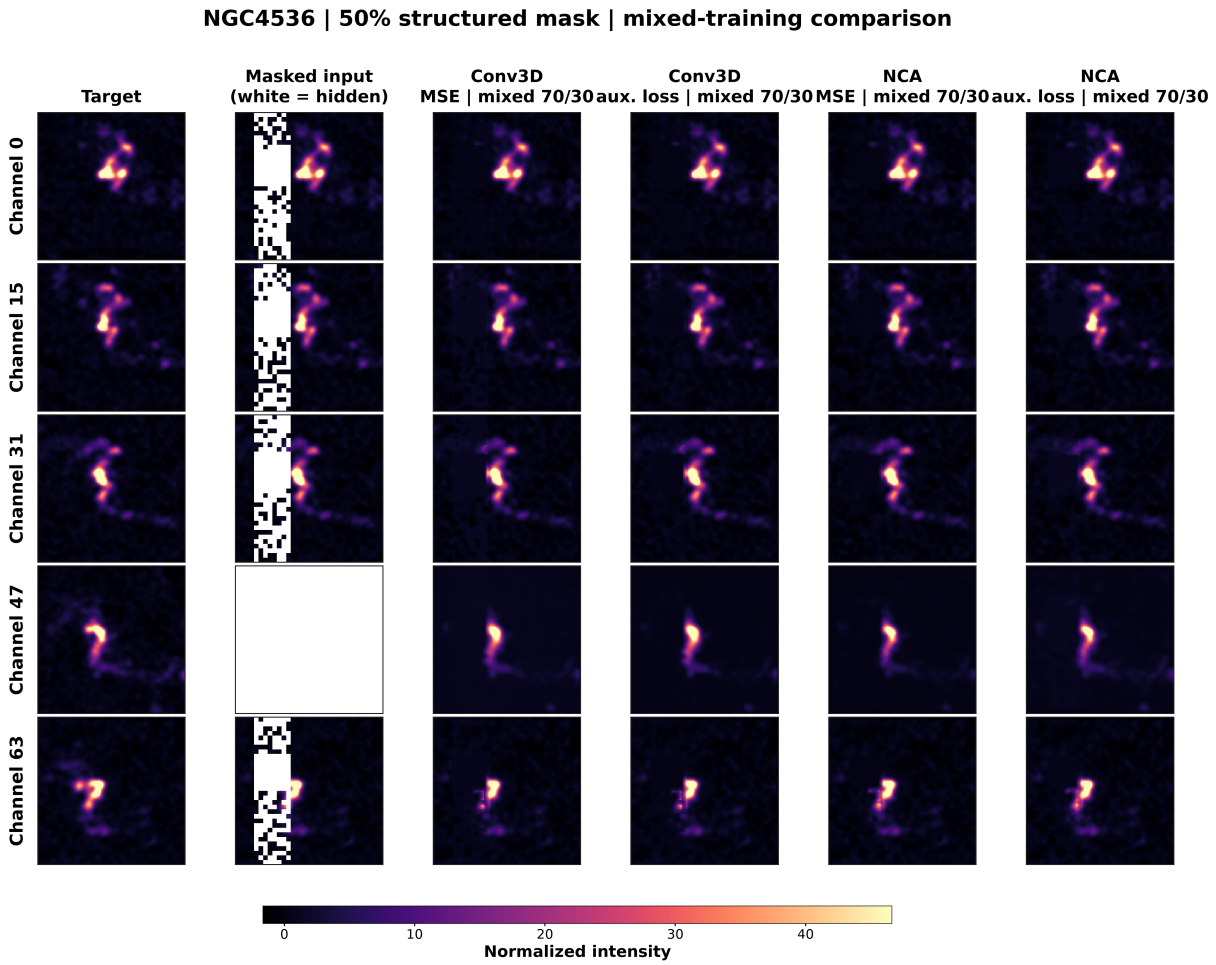


Figure 9: Structured-mask comparison of the four mixed-training configurations. The target and masked input precede the directly labelled Conv3D and NCA reconstructions; white denotes hidden input and every panel uses the same robust intensity scale.

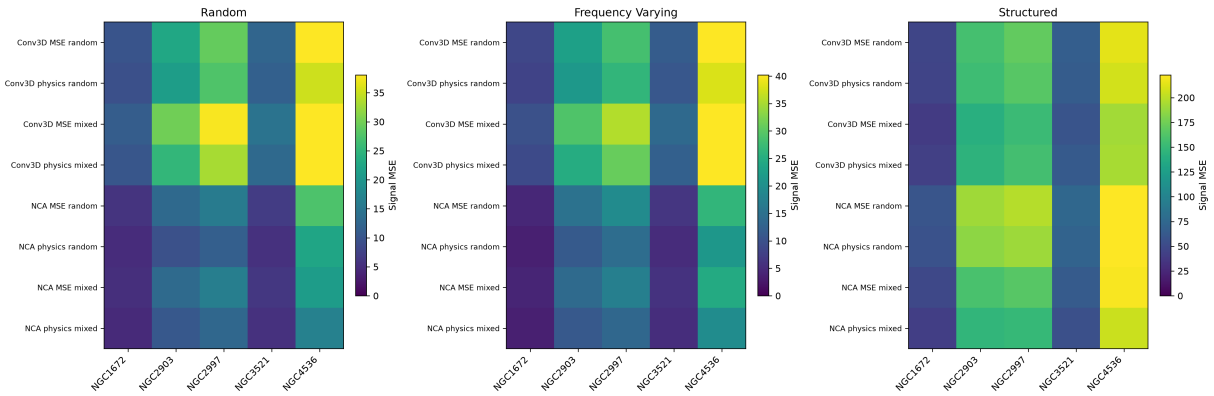


Figure 10: Per-galaxy signal-region MSE across the final factorial model set and evaluation masks.